HW11 Report

DATA 266 — Homework 11

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SID4 = 0670 | SEED = 670 | SLICE = 670 | HP_ID = 4 | CLS_A = 0 | CLS_B = 5

Notebook: `clip_cifar.ipynb`. Model: `openai/clip-vit-base-patch32`. Dataset: CIFAR-10.

HP_ID is reported only.

1. Setup

Item	Value
Model	`openai/clip-vit-base-patch32` (~151M params)
Dataset	CIFAR-10
Caption	`a photo of a dog`
Part A sample	100 images per class (SEED=670)
Zero-shot prompts	`a photo of a {class}`
Device	MPS
Batch	64

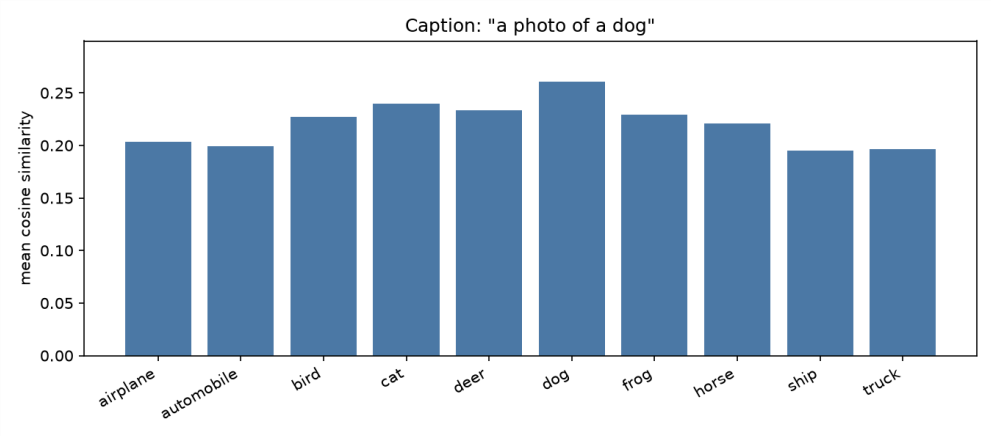
I load CLIP with the matching processor. Text and image embeddings are L2-normalized, then compared with cosine similarity (dot product of unit vectors).

2. Caption–image similarity

I scored the caption against 1,000 CIFAR-10 train images (100 per class).

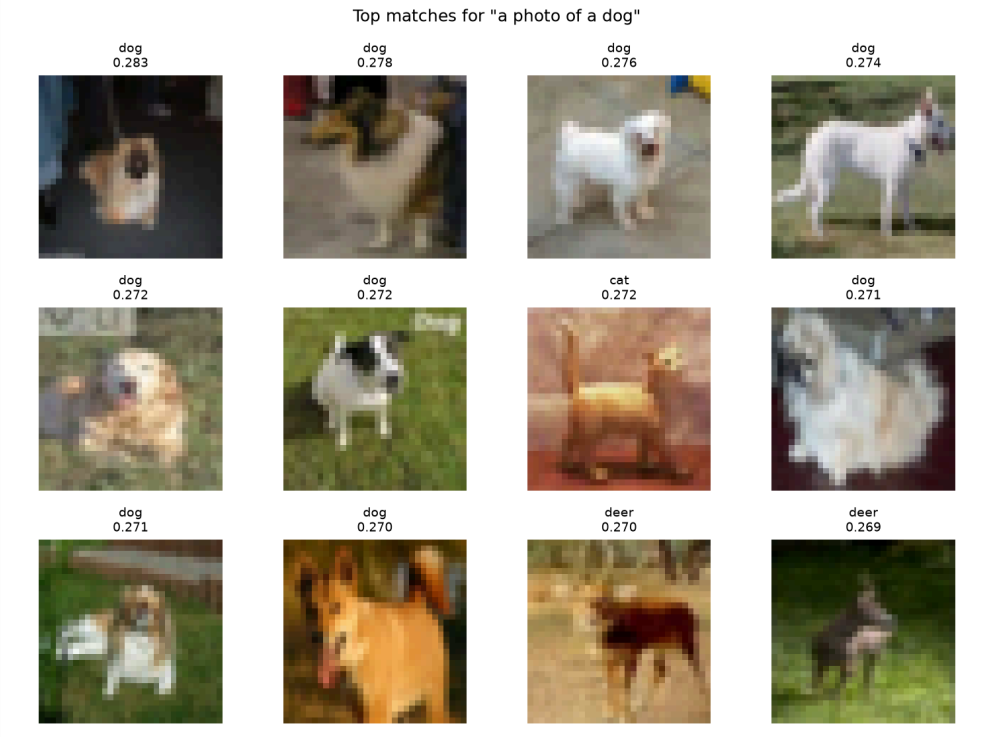
Class	Mean similarity
airplane	0.204

Class	Mean similarity
automobile	0.200
bird	0.227
cat	0.240
deer	0.234
dog	**0.260**
frog	0.229
horse	0.221
ship	0.195
truck	0.196



Mean caption similarity by class

Dog has the highest mean score (0.260). Cat is next (0.240), then other animals. Vehicles sit at the bottom. That is the expected ranking for a dog caption.



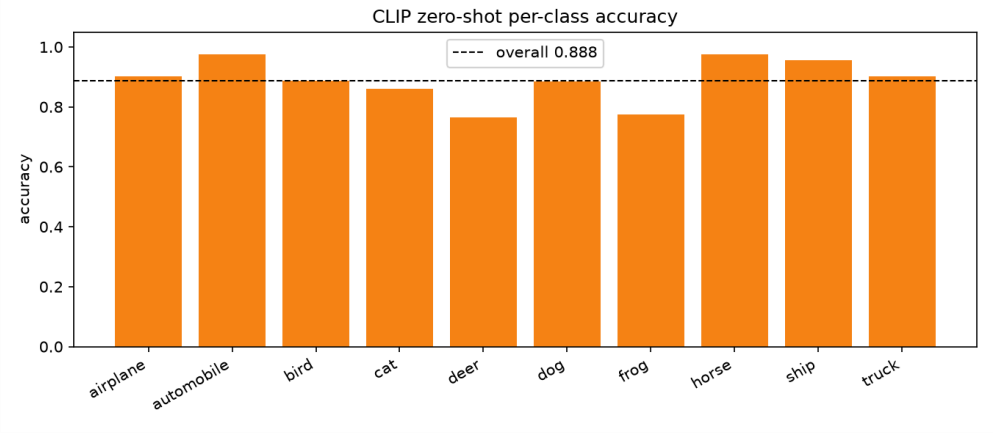
Top matches for the caption

The top-12 gallery is mostly dog images, with a few other animals that still look somewhat dog-like at 32×32.

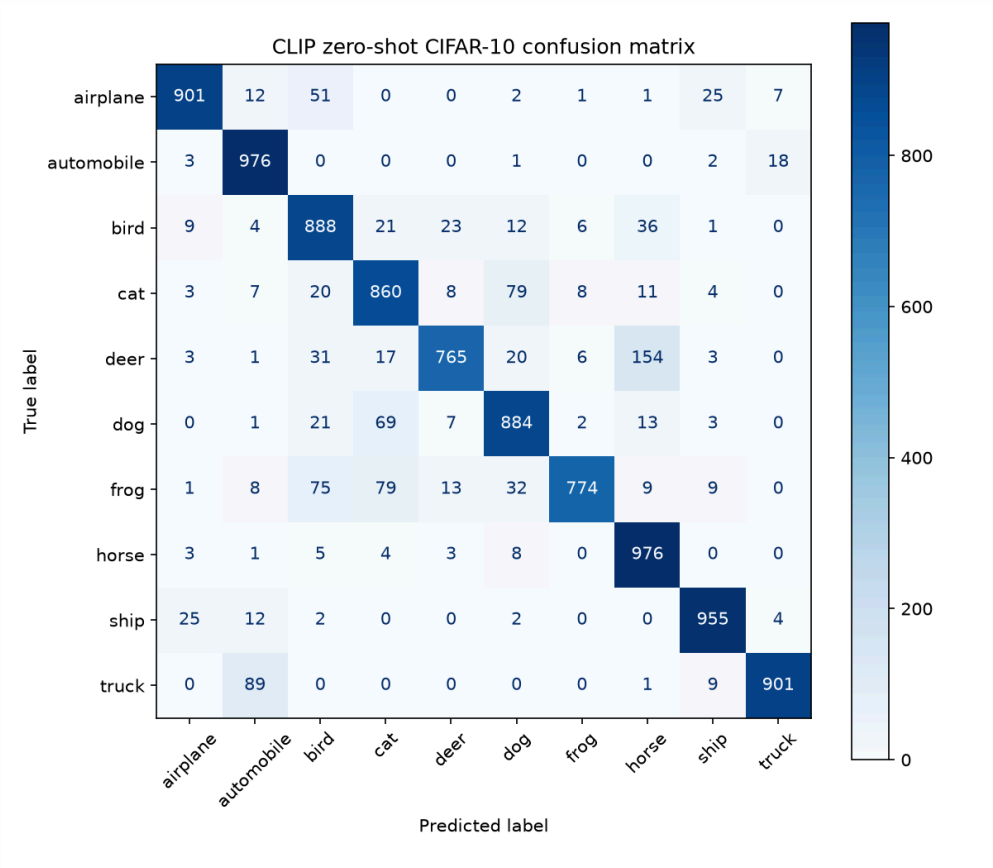
3. Zero-shot CIFAR-10 classification

I embedded all 10 class prompts and classified every CIFAR-10 test image by the highest cosine similarity. Overall accuracy: **0.888**.

Class	Accuracy
airplane	0.901
automobile	0.976
bird	0.888
cat	0.860
deer	0.765
dog	0.884
frog	0.774
horse	0.976
ship	0.955
truck	0.901



Per-class zero-shot accuracy



Confusion matrix

Automobile and horse are strongest (97.6%). Deer is weakest (76.5%), with many deer predicted as horse. Cat↔dog and frog↔bird/cat swaps show up in the matrix; those classes share texture and shape at CIFAR resolution.

Conclusion

CLIP ranked the dog caption highest on dog images without any fine-tuning. Zero-shot CIFAR-10 reached 88.8% accuracy. Most errors are between visually similar classes (deer/horse, cat/dog), which matches what the confusion matrix

shows.